

($n-3$)-Edge-Fault-Tolerant Weak-Pancyclicity of (n, k) -Star Graphs

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Abstract

The (n, k) -star graph is a generalized version of the n -star graph, which belongs to the class of Cayley graphs, and has been recognized as an attractive alternative to an n -cube for building massively parallel computers. Recently, Chen et al. showed that an (n, k) -star graph is 6-weak-vertex-pancyclic for $k < n-1$, that is, each vertex of an (n, k) -star graph is contained in a cycle of length ranged from 6 to $n!/(n-k)!$. This work demonstrates that an (n, k) -star graph remains 6-weak-pancyclic, even if there are up to $n-3$ edge faults, where $n \geq 4$ and $1 \leq k < n$. Since an (n, k) -star graph is regular of degree $n-1$, the result of this work is optimal with respect to the number of edge faults tolerated.

1 Introduction

Recently, multiprocessors can be built in a single-chip [19]. Constructing a massively parallel computer system containing thousands or even more processors is highly possible in near future [21]. However, to optimize the communication efficiency of a multiprocessor system, the interconnection network in it should be investigated, and this problem can be modeled by a graph problem [24]. Graph theory is very useful for designing and analyzing topological properties of interconnection networks. The n -star graph (S_n for short), which proposed by Akers et al. in 1987, is an attractive alternative to the hypercube with vertex and edge symmetry [1]. An (n, k) -star graph ($S_{n,k}$ for short) refers to a generalized version of an S_n , where $S_{n,n-1}$ and S_n are isomorphic and $S_{n,1}$ is obviously a complete graph of n vertices (K_n) [7]. An $S_{n,k}$ is regular of degree $n-1$ and has $n!/(n-k)!$ vertices. An $S_{n,k}$ preserves many attractive properties of an S_n such as vertex symmetry, hierarchical structure, maximal fault tolerance, and simple shortest routing [7]. Some basic properties of an $S_{n,k}$, such as, diameter [7], connectivity [7], broadcasting [5], average

distance [8], embedding [3], [4], Hamiltonicity [13], spanning connectivity [14], and wide diameter [16], have recently been computed or derived. These measurement results demonstrate that an $S_{n,k}$ has excellent topological properties.

Cycle is one of the most fundamental networks for parallel and distributed computation, and very suitable for designing simple algorithms with low communication costs. Each parallel algorithm for a cycle can be applied to a network which contains the cycle. Therefore, cycle embedding is very important issue for networks and is widely discussed in many researches [2], [3], [4], [6], [9], [10], [11], [12], [15], [17], [18], [22], [23], [25]. A graph G of order $|V(G)|$ is *pancyclic* whenever G contains a cycle of length ranged from 3 to $|V(G)|$ [2]. Various generalizations of pancyclic graphs have been studied, such as Hypercubes [17], [20], Möbius cubes [3], [23], 3-ary N -cubes [12], bubble-sort graphs [15], twisted cubes [10], [11], crossed cubes [9], pyramids [21], and n -star graphs [18], [25]. A graph G is *f -edge-fault-tolerant pancyclic* if $G - F_e$ remains pancyclicity for every $F_e \subset E(G)$ with $|F_e| \leq f$. An *f -edge-fault-tolerant* graph G is a graph that still has some property by removing any f edges in G . Additionally, a graph G is *vertex-pancyclic* (*vertex-bipancyclic*) if every vertex lies on a cycle of every length (every even length) from 3 (4) to $|V(G)|$. Similarly, a graph G is *edge-pancyclic* (*edge-bipancyclic*) if every edge lies on a cycle of every length (every even length) from 3 (4) to $|V(G)|$. A graph G is *bipancyclic* if it has a cycle of every even length ranged from 4 to $|V(G)|$. An *m -weak-pancyclic* (*m -weak-bipancyclic*) graph is a graph which contains a cycle of every length (every even length) l for $m \leq l \leq |V(G)|$. Furthermore, a graph G is *m -weak-vertex-pancyclic* (*m -weak-vertex-bipancyclic*) if every vertex lies on a cycle of every length (every even length) from m to $|V(G)|$. Let $d(u, v)$ represents the distance between two vertices u and v in a graph G . A graph G is *m -weak-panconnected* if there exists a path of length l between each pair of distinct vertices u and v for $\max\{m, d(u, v)\} \leq l \leq |V(G)|-1$.

Recently, Li already showed that an S_n contains a cycle of every even length from 6 to $n!$ under

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$n-3$ faulty edges [18]. Xu et al. have shown that an S_n (or $S_{n,n-1}$) is $(n-3)$ -edge-fault-tolerant 6-weak-edge-bipancyclic [24]. Moreover, Chen et al. demonstrate in 2008 that an $S_{n,k}$ is 6-weak-vertex-pancyclic for $k \leq n-2$ [6]. Since Edge fault tolerance is also an important issue for parallel computers, this work further investigates that an $S_{n,k}$ remains weak-pancyclic, even if there are up to $n-3$ edge faults, where $n \geq 4$ and $1 \leq k < n$.

The remainder of this paper is organized as follows. Section 2 formally defines the structure of an $S_{n,k}$ in terms of a graph. Section 3 demonstrates how to embed cycles in an $S_{n,k}$ graph under edge faults. Conclusions are finally drawn in Section 4.

2 Background and Notations

An $S_{n,k}$ is a Cayley graph with a regular and hierarchical structure, and has been recognized as an attractive alternative to an n -cube for building massively parallel computers [7]. This section formally presents the structure of the $S_{n,k}$ and studies some basic properties of it.

For simplicity, let $\langle n \rangle = \{1, 2, \dots, n\}$, $\langle k \rangle = \{1, 2, \dots, k\}$ and two positive integers n and k satisfy $1 \leq k \leq n-1$. Parameters n and k determine the topological properties of an $S_{n,k}$, such as number of vertices, connectivity and diameter [7]. The vertex set of an $S_{n,k}$ $V(S_{n,k}) = \{p_1 p_2 \dots p_k \mid p_i \in \langle n \rangle \text{ and } p_i \neq p_j \text{ for } i \neq j\}$ and the edge set of an $S_{n,k}$ $E(S_{n,k}) = \{(p_1 p_2 \dots p_{i-1} p_i p_{i+1} \dots p_k, p_i p_2 \dots p_{i-1} p_1 p_{i+1} \dots p_k), (p_1 p_2 \dots p_k, \alpha p_2 \dots p_k) \mid p_1 p_2 \dots p_k \in V(S_{n,k}) \text{ and } \alpha \in \langle n \rangle - \{p_q \mid 1 \leq q \leq k\}, 2 \leq i \leq k\}$. Notably, each vertex of an $S_{n,k}$ is represented by a k -symbol string and each symbol belongs to $\langle n \rangle$.

Edge $(p_1 p_2 \dots p_{i-1} p_i p_{i+1} \dots p_k, p_i p_2 \dots p_{i-1} p_1 p_{i+1} \dots p_k)$ is referred to as i -edge and the two endvertices are i -neighbors to each other, and each edge $(p_1 p_2 \dots p_k, \alpha p_2 \dots p_k)$ is referred to as a 1-edge and the two endvertices of it are 1-neighbor to each other [7], [16]. The structure of an $S_{4,2}$ is shown in Figure 1.

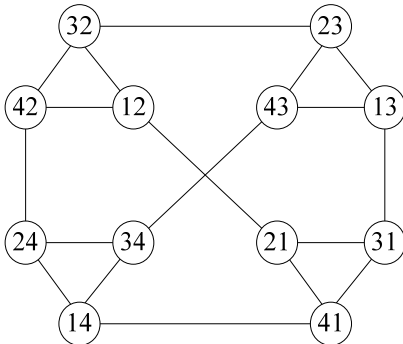


Figure 1. The structure of an $S_{4,2}$.

Let $S_{n-1,k-1}(\omega)$ denote a subgraph of an $S_{n,k}$, which induced by all the vertices with the same last symbol ω , for some $1 \leq \omega \leq n$. By the structure of the (n, k) -star graph, 1-edges and i -edges still remain what they are in an $S_{n-j,k-j}$ for $1 \leq j \leq k-1$ and $1 \leq i \leq k-j$.

An $S_{n,k}$ can be formed by interconnecting n $S_{n-1,k-1}$'s. Figure 1 indicates that an $S_{4,2}$ can be viewed as an interconnection of $S_{3,1}(\omega)$'s, $1 \leq \omega \leq 4$, through 2-edges. That is, an $S_{n,k}$ can be decomposed into $S_{n-1,k-1}$'s along any dimension i , and it can also be decomposed into n vertex-disjoint $S_{n-1,k-1}$'s in $k-1$ different ways by fixing the symbol in any position i , $2 \leq i \leq k$ [7], [8]. Moreover, In an $S_{n,k}$, every 1-edge is edge-symmetric with any other 1-edge [8]; every i -edge is edge-symmetric with any other i -edge [8], and the length of a cycle containing an i -edge is at least 6, where $2 \leq i \leq k$ [16].

A cycle (path) in a graph G is a Hamiltonian cycle (Hamiltonian path) if it contains every vertex of G exactly once. A graph G is Hamiltonian if it has a Hamiltonian cycle, and a graph G is Hamiltonian-connected if it has a Hamiltonian path between every two of its vertices. Let $F_v \subset V(G)$ and $F_e \subset E(G)$. If $G - F_v - F_e$ is Hamiltonian (Hamiltonian-connected), then G is called f -fault-tolerant Hamiltonian (f -fault-tolerant Hamiltonian-connected) as long as $|F_v| + |F_e| \leq f$.

Lemma 1 [13]. An $S_{n,k}$ is $(n-3)$ -fault-tolerant Hamiltonian and $(n-4)$ -fault-tolerant Hamiltonian connected if $n-k \geq 2$.

Theorem 2 [6]. An $S_{n,k}$ is edge-pancyclic, if $k = 1$ and $n \geq 3$, vertex-pancyclic, if $2 \leq k \leq n-4$ and $n \geq 6$, 6-weak-vertex-pancyclic, if $\max\{2, n-3\} \leq k \leq n-2$ and $n \geq 4$, 6-weak-edge-bipancyclic, if $k = n-1$ and $n \geq 3$. Moreover, each cycle of an available length in an $S_{n,k}$ can contain a desired 1-edge.

3 Cycle Embedding under Edge Faults

Since Xu et al. have already shown that an $S_{n,n-1}$ (S_n) is $(n-3)$ -edge-fault-tolerant 6-weak-edge-bipancyclic [24], an $S_{n,n-1}$ is $(n-3)$ -edge-fault-tolerant 6-weak-bipancyclic. This section first shows that $S_{n,1}$ and $S_{n,2}$ are $(n-3)$ -edge-fault-tolerant pancyclic and then proves an $S_{n,k}$ is $(n-3)$ -edge-fault-tolerant pancyclic by induction on k and n for $3 \leq k \leq n-4$ and $n \geq 7$. Finally, the two special cases of $S_{n,n-2}$ and $S_{n,n-3}$ are also discussed.

Lemma 3. An $S_{4,1}$ is 1-edge-fault-tolerant pancyclic.

Proof. Since an $S_{4,1}$ is isomorphic to K_4 and edge-symmetric, assume that $F_e = \{(1,3)\}$ and the resulting graph as shown in the Figure 2. Then the cycles of lengths 3 and 4 can be constructed as $\langle 1, 2, 4, 1 \rangle$ and $\langle 1, 2, 3, 4, 1 \rangle$, respectively. Hence, the lemma follows. $\square$

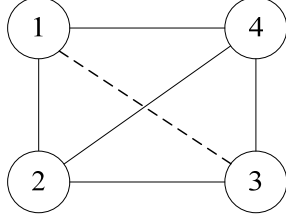


Figure 2. An $S_{4,1}$ with faulty edge $(1,3)$.

Lemma 4. An $S_{n,1}$ is $(n-3)$ -edge-fault-tolerant pancyclic for $n \geq 5$.

Proof. This lemma is proved by induction on n . Lemma 3 indicates that an $S_{4,1}$ is $(n-3)$ -edge-fault-tolerant pancyclic. Assume that the statement is true for all $4 \leq k \leq n-1$. Let p, q be two vertices in the $S_{n,1}$ and $(p, q) \in F_e$. Since $|F_e| \leq n-3$, the induced subgraph $S_{n,1} - \{p\}$ is a K_{n-1} and contains at most $n-4$ faulty edges. By induction hypothesis, there exists a cycle of length ranged from 3 to $n-1$ in the graph $S_{n,1} - \{p\}$. According to Lemma 1, an $S_{n,1}$ is $(n-3)$ -fault-Hamiltonian. Hence, the lemma follows. $\square$

Lemma 5. An $S_{n,2}$ is $(n-3)$ -edge-fault-tolerant pancyclic for $n \geq 6$.

Proof. An $S_{n,2}$ is composed of n $S_{n-1,1}(\omega)$'s where $1 \leq \omega \leq n$. Since $|F_e| \leq n-3$, there exist at least three subgraphs $S_{n-1,1}(\omega)$'s, which are fault-free and have a cycle of length ranged from 3 to $n-1$. Three cases are needed to be considered for a cycle of length ranged from n to $n(n-1)$, which is contained in at least three $S_{n-1,1}(\omega)$'s. Regarding each $S_{n-1,1}(\omega)$ of the $S_{n,2}$ as a supervertex, the resulting graph is an $S_{n,1}$. By Lemma 4, an $S_{n,2}$ with $|F_e| \leq n-3$ contains a cycle $\langle u_1, v_1, u_2, v_2, \dots, u_{l/2}, v_{l/2}, u_1 \rangle$ of even length l for $6 \leq l \leq 2n$, denoted by σ_l , where 1-edge $(u_i, v_i) \in E(S_{n-1,1}(a_i))$ and $1 \leq i \leq l/2$, $1 \leq a_i \leq n$.

Case 1. Every faulty edge is 2-edge. A cycle of length ranged from n to $n(n-1)$ is built step by step as follows.

1. Construct $\sigma_6 = \langle u_1, v_1, u_2, v_2, u_3, v_3, u_1 \rangle$ as the basic cycle to expand. According to Theorem 2, a cycle of length at most $n-1$ can be constructed in the $S_{n-1,1}(a_i)$ and the cycle can contain any 1-edge, where $1 \leq i \leq 3$. Hence, a cycle of length ranged from 3 to $n-1$ could be easily

constructed in a $S_{n-1,1}(a_i)$ including the specified 1-edge. Building a cycle of length $n-4$ with the 1-edge (u_1, v_1) in the $S_{n-1,1}(a_1)$, a cycle of length n can be obtained after eliminating the 1-edge (u_1, v_1) as shown in Figure 3.

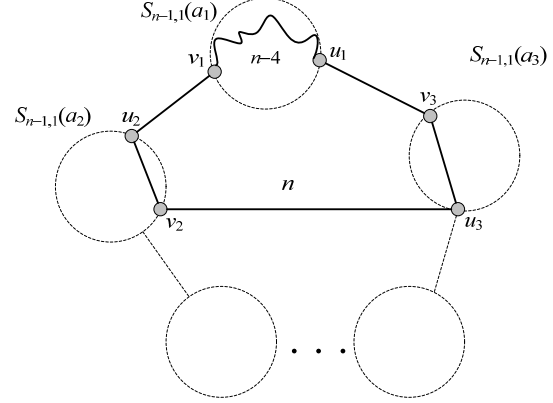


Figure 3. A cycle of length n in an $S_{n,2}$.

2. By adding vertices in these three $S_{n-1,1}(a_i)$'s, which are excluded by the cycle constructed in Step 1, into the cycle one-by-one, a cycle of length ranged from $n+1$ to $3n-3$ can be established. Figure 4 illustrates the cycle of length $3n-3$.

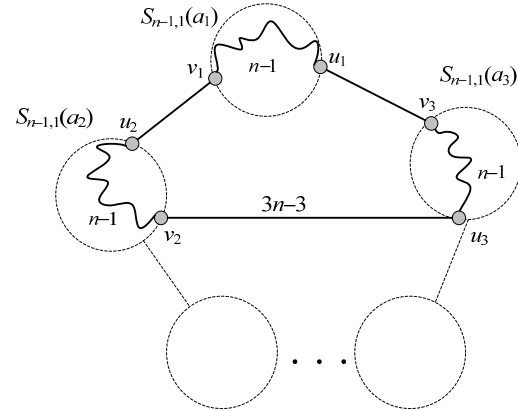


Figure 4. A cycle of length $3n-3$ in an $S_{n,2}$.

3. For constructing a cycle of length $3n-2$, adopt a cycle σ_8 in the $S_{n,2}$ including 4 $S_{n-1,1}(a_i)$'s as the basic cycle, where $1 \leq i \leq 4$. Then build a cycle of length $n-1$ with the 1-edge (u_j, v_j) in each $S_{n-1,1}(a_i)$ for $1 \leq j \leq 2$ and build a cycle of length $n-2$ with the 1-edge (u_3, v_3) in the $S_{n-1,1}(a_3)$. Significantly, vertices u_3 and v_2 are not the same as in Steps 1 and 2. A cycle of length $3n-2$ can be constructed after removing 1-edges (u_1, v_1) , (u_2, v_2) and (u_3, v_3) as shown in Figure 5. Referring to Step 2, a cycle of length ranged from $3n-1$ to $4n-4$ can be established.
4. According to the cycle construction in Step 3, a cycle of length ranged from $4n-3$ to $n(n-1)$ can

be built.

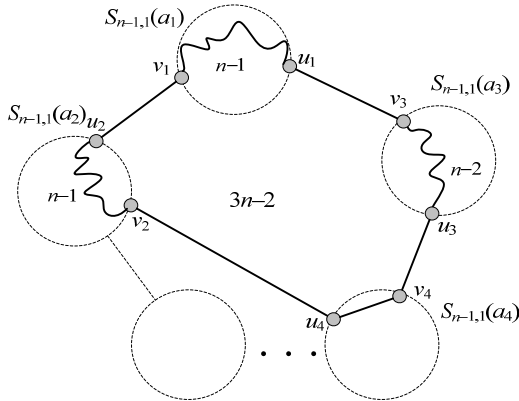


Figure 5. A cycle of length $3n-2$ in an $S_{n,2}$.

Case 2. Every faulty edge is 1-edge. Without loss of generality, assume that $|F_e \cap E(S_{n-1,1}(a_n))| \geq |F_e \cap E(S_{n-1,1}(a_{n-1}))| \geq \dots \geq |F_e \cap E(S_{n-1,1}(a_1))|$ for $1 \leq a_i \leq n$. Hence, $|F_e \cap E(S_{n-1,1}(a_n))| \leq n-3$ and $|F_e \cap E(S_{n-1,1}(a_i))| \leq n-5$, where $1 \leq i \leq n-1$. Otherwise, there exists a subgraph $S_{n-1,1}(a_i)$ such that $|F_e \cap E(S_{n-1,1}(a_i))| = n-4$ where $1 \leq i \leq n-1$. Since $|F_e \cap E(S_{n-1,1}(a_i))| \leq |F_e \cap E(S_{n-1,1}(a_n))|$, $|F_e| \geq 2(n-4) > n-3$ for $n \geq 6$ and it is a contradiction. Because $|F_e| \leq n-3$, assume that subgraphs $S_{n-1,1}(a_1)$, $S_{n-1,1}(a_2)$ and $S_{n-1,1}(a_3)$ are fault-free. According to Case 1, a cycle of length ranged from n to $3n-3$ can be easily constructed. Significantly, $|F_e \cap E(S_{n-1,1}(r))| \leq n-5$. A cycle of length ranged from $3n-2$ to $n(n-1)$ is build step by step as follows.

1. Let $\sigma_6 = \langle u_1, v_1, u_2, v_2, u_3, v_3, u_1 \rangle$ be the cycle of length 6, where 1-edge $(u_i, v_i) \in E(S_{n-1,1}(a_i))$ and $1 \leq i \leq 3$. A cycle of length $3n-2$ can be constructed by including an $S_{n-1,1}(a_4)$ where $a_4 \in R$. Let (v'_3, u_4) and (u'_1, v_4) be 2-edges where $v'_3 \in V(S_{n-1,1}(a_3))$, $u'_1 \in V(S_{n-1,1}(a_1))$ and $u_4, v_4 \in V(S_{n-1,1}(a_4))$. Construct a cycle of length $n-2$ with 1-edge (u'_1, v_1) in the $S_{n-1,1}(a_1)$, build a cycle of length $n-1$ with 1-edge (u_3, v'_3) in the $S_{n-1,1}(a_3)$. According to Lemma 1, there exists a Hamiltonian path between u_4 and v_4 in the $S_{n-1,1}(a_4)$. A cycle of length $3n-2$ can be obtained after eliminating 1-edges (u'_1, v_1) and (u_3, v'_3) as shown in Figure 6.
2. By adding vertices in $S_{n-1,1}(a_1)$ and $S_{n-1,1}(a_2)$, which are excluded by the cycle constructed in Step 1, into the cycle one-by-one, a cycle of length ranged from $3n-1$ to $4n-4$ can be established.
3. Similarly, referring to Steps 1 and 2, a cycle of length ranged from $4n-3$ to $(n-1)(n-1)$ can be built by adding 2 to $n-1$ $S_{n-1,1}(a_j)$'s one by one, where $5 \leq j \leq n$.
4. According to Lemma 1, there exists a Hamiltonian cycle $HC = \langle u_1, \dots, v_1, u_2, \dots, v_2, \dots,$

$u_n, \dots, v_n, u_1 \rangle$ in an $S_{n,2}$ with vertices $u_i, v_i \in V(S_{n-1,1}(a_i))$ and each subpath $\langle u_i, \dots, v_i \rangle$ in HC is a Hamiltonian path of the $S_{n-1,1}(a_i)$, where $1 \leq i \leq n$.

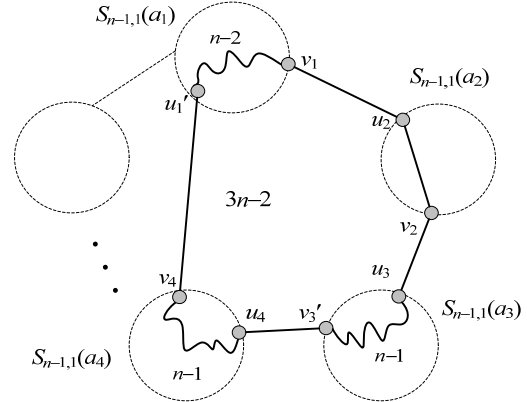


Figure 6. A cycle of length $3n-2$ in an $S_{n,2}$.

5. For constructing a cycle of length $(n-1)(n-1)+1$, construct a cycle of length $n-2$ with 1-edge (u_1, v_1) in the $S_{n-1,1}(a_1)$, and replace the subpath $\langle u_2, \dots, v_2 \rangle$ in HC by edge (u_2, v_2) . Thus, a cycle of length $(n-1)(n-1)+1$ can be obtained after eliminating 1-edge (u_1, v_1) as shown in Figure 7.

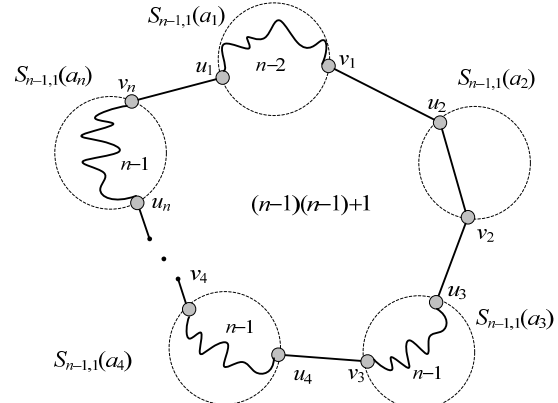


Figure 7. A cycle of length $(n-1)(n-1)+1$ in an $S_{n,2}$.

6. By adding vertices in $S_{n-1,1}(a_1)$ and $S_{n-1,1}(a_2)$, which are excluded by the cycle constructed in Step 5, into the cycle one-by-one, a cycle of length ranged from $(n-1)(n-1)+2$ to $n(n-1)-1$ can be established.

Case 3. Otherwise. Combining the cycle constructions in Cases 1 and 2, a cycle of length ranged from n to $n(n-1)$ can be built. $\square$

An $S_{4,2}$ ($S_{5,2}$) is composed of 4 $S_{3,1}$'s (5 $S_{4,1}$'s) and has a cycle of length l for $l = 3$ ($l = 3, 4$). Moreover, Lin and Duh showed that the length of a cycle containing an i -edge is at least 6, where 2

$\leq i \leq k$ [16]. Therefore, the $S_{4,2}$ ($S_{5,2}$) has no cycle of length 4 (4 and 5). With the aid of computer simulation, $S_{4,2}$ and $S_{5,2}$ are both $(n-3)$ -edge-fault-tolerant 6-weak-pancyclic.

In the following, $(n-3)$ -edge-fault-tolerant pancyclicity of an $S_{n,k}$ is proved by cycle construction for $3 \leq k \leq n-4$ and $n \geq 7$. Additionally, $S_{n,n-2}$ and $S_{n,n-3}$ are special cases and shown individually. For the cycle construction, $n-2$ vertex-disjoint cycles of length 6 and $n-2$ vertex-disjoint paths crossing the given number of subgraphs are needed to be constructed to guarantee that at least one of these cycles and paths are fault-free. After combining the fault-free cycle and path as basis, cycles of required lengths can be constructed by including vertex one by one.

Two algorithms are derived in the following. Algorithm 1 establishes $n-2$ vertex-disjoint cycles of lengths 6 crossing three subgraphs and each cycle consists of 1-edges and k -edges. Algorithm 2 constructs $n-2$ vertex-disjoint paths crossing m subgraphs for $3 \leq m \leq n$.

Let $p \rightarrow_{(j)} q$ denote that vertices p and q are connected by a j -edge where $1 \leq j \leq k$. Also let p_i denote the i th digits of vertex p .

Algorithm 1

Input: An $S_{n,k}$ for $n \geq 4$, $k \geq 3$ and two distinct numbers p_1 and p_k with $p_1, p_k \in \langle n \rangle$.

Output: $n-2$ vertex-disjoint cycles of lengths 6, denoted by $C^1, C^2, \dots$, and C^{n-2} , respectively.

1. Select $n-2$ distinct vertices $p^i = p_1^i p_2^i \dots p_k^i$ for $1 \leq i \leq n-2$ from $V(S_{n-1,k-1}(p_k))$ such that $p_1^i = p_1$ and $p_k^i = p_k$.
2. Let $\alpha^i \in \langle n \rangle - \{p_q^i \mid 1 \leq q \leq k\}$ where $1 \leq i \leq n-2$.
3. For $1 \leq i \leq n-2$, C^i is constructed as $p^i \rightarrow_{(k)} p_k^i p_2^i p_3^i \dots p_{k-1}^i p_1^i \rightarrow_{(1)} \alpha^i p_2^i p_3^i \dots p_{k-1}^i p_1^i \rightarrow_{(k)} p_1^i p_2^i p_3^i \dots p_{k-1}^i \alpha^i \rightarrow_{(1)} p_1^i p_2^i p_3^i \dots p_{k-1}^i \alpha^i \rightarrow_{(k)} \alpha^i p_2^i p_3^i \dots p_{k-1}^i p_1^i \rightarrow_{(1)} p^i$.
4. Output $C^1, C^2, \dots$, and C^{n-2} .

Each cycle constructed by algorithm 1 contains two vertices in each of $S_{n-1,k-1}(p_k)$, $S_{n-1,k-1}(p_1)$ and $S_{n-1,k-1}(\alpha^i)$, and each edge in each cycle is either 1-edge or k -edge. If $|F_e| \leq n-3$, at least one of these constructed cycles is fault-free. Example 1 demonstrates 4 vertex-disjoint cycles in an $S_{6,4}$ constructed by Algorithm 1 and Lemma 6 shows that cycles constructed by algorithm 1 are vertex-disjoint to each other.

Example 1. In an $S_{6,4}$, let $p^1 = 1234$, $p^2 = 1254$, $p^3 = 1264$, $p^4 = 1654$, $\alpha^1 = 6$, $\alpha^2 = 6$, $\alpha^3 = 5$, and $\alpha^4 = 3$. For $1 \leq i \leq n-2$, C^i can be constructed as follows.

- C^1 : $\langle 1234, 4231, 6231, 1236, 4236, 6234, 1234 \rangle$,
 C^2 : $\langle 1254, 4251, 6251, 1256, 4256, 6254, 1254 \rangle$,
 C^3 : $\langle 1264, 4261, 5261, 1265, 4265, 5264, 1264 \rangle$,

C^4 : $\langle 1654, 4651, 3651, 1653, 4653, 3654, 1654 \rangle$.

Lemma 6. Cycles $C^1, C^2, \dots, C^{n-2}$ constructed by Algorithm 1 are vertex-disjoint.

Proof. According to Step 3 in Algorithm 1, every vertex in C^i is distinct, where $1 \leq i \leq n-2$. Moreover, every vertex in C^i is disjoint to every vertex in C^j where $i \neq j$ and $1 \leq i, j \leq n-2$, because $p_2^i p_3^i \dots p_{k-1}^i \neq p_2^j p_3^j \dots p_{k-1}^j$. Therefore, $C^1, C^2, \dots$, and C^{n-2} are vertex-disjoint. $\square$

Significantly, one of cycles constructed by Algorithm 1 is fault-free. Before providing Algorithm 2 to construct $n-2$ paths, Lemmas 7 and 8 are given for proving the paths constructed by Algorithm 2 are vertex-disjoint to each other.

Lemma 7. Let u and v be two distinct vertices in an $S_{n,k}$ for $n \geq 3$ and $2 \leq k \leq n-1$. If $u_1 = v_1$ then $d(u, v) \geq 3$.

Proof. According to the structure of an $S_{n,k}$, at least one digit of u and v is different. Assume that $u_i \neq v_i$, where $2 \leq i \leq k$. The shortest path between u and v should be constructed as $u = u_1 u_2 u_3 \dots u_k \rightarrow_{(1)} v_i u_2 u_3 \dots u_k \rightarrow_{(i)} u_1 u_2 u_3 \dots u_{i-1} v_i u_{i+1} \dots u_k \rightarrow_{(1)} u_1 u_2 u_3 \dots u_{i-1} v_i u_{i+1} \dots u_k = v_1 v_2 v_3 \dots v_k = v$. Notably, $d(u, v)$ dependent on the number of misplaced digits [7]. Thus, $d(u, v) \geq 3$. $\square$

Lemma 8. Let $p^1, p^2, \dots, p^{n-2}$ be distinct vertices in an $S_{n-1,k-1}(\omega)$ where $1 \leq \omega \leq n$, $n \geq 3$, $2 \leq k \leq n-1$, and $p_1^1 = p_1^2 = \dots = p_1^{n-2}$. If u^i be a neighbor of p^j for $1 \leq j \leq n-2$, then p^j and u^i are distinct.

Proof. If $u^i = p^j$ ($p^i = u^j$), where $1 \leq i \neq j \leq n-2$, then $d(p^i, p^j) = 2$ ($d(p^i, p^j) = 1$). According to Lemma 7, $d(p^i, p^j) \geq 3$. Therefore, $u^i \neq p^j$ ($p^i \neq u^j$). $\square$

Let $p = p_1 p_2 \dots p_k$ be a vertex in the $S_{n-1,k-1}(p_k)$ and $\alpha \in \langle n \rangle - \{p_r \mid 1 \leq r \leq k\}$. Also, let P_j^i be a path with $2j+2$ vertices from $\gamma_1 p_2^i p_3^i \dots p_{k-2}^i p_{k-1}^i p_k^i$ to $\gamma_2 q_2^i q_3^i \dots q_{k-2}^i q_{k-1}^i \alpha$ containing 2 adjacent vertices in each $S_{n-1,k-1}(\gamma_j)$, where $\gamma_j \in \langle n \rangle - \{p_1, p_k, \alpha\}$, $p_s^i, q_s^i \in \langle n \rangle$, $2 \leq s \leq k-1$, $1 \leq j \leq n-3$, and $1 \leq i \leq n-2$. Without loss of generality, assume that P_1^i contains 2 adjacent vertices in the $S_{n-1,k-1}(\gamma_1)$; P_2^i contains 2 adjacent vertices in each of $S_{n-1,k-1}(\gamma_1)$ and $S_{n-1,k-1}(\gamma_2)$, and so on. The following algorithm demonstrates how to construct $n-2$ paths orderly crossing $S_{n-1,k-1}(p_k)$, $S_{n-1,k-1}(\gamma_1)$, $S_{n-1,k-1}(\gamma_2), \dots$, $S_{n-1,k-1}(\gamma_j)$, $S_{n-1,k-1}(\alpha)$ in an $S_{n,k}$ for $n \geq 4$, $k \geq 3$ and then shows the constructed paths are vertex-disjoint.

Algorithm 2

Input: Vertices $p = p_1 p_2 \dots p_k$ and $\gamma_1 p_2^i p_3^i \dots p_{k-2}^i p_{k-1}^i p_k^i$ for $1 \leq i \leq n-2$, $\alpha \in \langle n \rangle - \{p_r \mid 1 \leq r \leq k\}$, and j represents the number of

$S_{n-1,k-1}(\omega)$ included, where $\omega \in \langle n \rangle - \{p_1, p_k, \alpha\}$.

Output: $n-2$ vertex-disjoint paths from $\gamma_1 p_2^i p_3^i \dots p_{k-2}^i p_{k-1}^i p_k$ to $\gamma_j q_2^i q_3^i \dots q_{k-2}^i q_{k-1}^i \alpha$, denoted by P_j^i for $1 \leq i \leq n-2$.

1. For $1 \leq i \leq n-2$, construct P_j^i from $\gamma_1 p_2^i p_3^i \dots p_{k-2}^i p_{k-1}^i p_k$ to $\gamma_j q_2^i q_3^i \dots q_{k-2}^i q_{k-1}^i \alpha$ containing 2 adjacent vertices in each $S_{n-1,k-1}(\gamma_j)$, where $\gamma_j \in \langle n \rangle - \{p_1, p_k, \alpha\}$, $p_s^i, q_s^i \in \langle n \rangle$, $2 \leq s \leq k-1$, $1 \leq j \leq n-3$, and $1 \leq i \leq n-2$.
2. Output P_j^i for $1 \leq i \leq n-2$.

Example 2 demonstrates 4 vertex-disjoint paths in an $S_{6,4}$ constructed by Algorithm 2.

Example 2. Let $p = 6413$ be a vertex in the $S_{5,3}(3)$ of an $S_{6,4}$. Choose 4 vertices $p^1 = 2143$, $p^2 = 2453$, $p^3 = 2163$ and $p^4 = 2653$ and then apply Algorithm 2 to construct 4 vertex-disjoint paths crossing $S_{5,3}(3)$, $S_{5,3}(2)$, $S_{5,3}(4)$, $S_{5,3}(1)$ as follows.

- $P_1^1: \langle 2143, 3142, 4132, 2134, 1234, 4231 \rangle$,
- $P_2^2: \langle 2453, 3452, 4352, 2354, 1354, 4351 \rangle$,
- $P_3^3: \langle 2163, 3162, 4162, 2164, 1264, 4261 \rangle$,
- $P_4^4: \langle 2653, 3652, 4652, 2654, 1654, 4651 \rangle$.

According to Lemma 8, it is easy to check that $n-2$ paths constructed by Algorithm 2 are vertex-disjoint. Naturally, one of the constructed $n-2$ paths is fault-free, since at most $n-3$ edges in an $S_{n,k}$ is faulty.

In the following, without loss of generality, assume that $|F_e \cap E(S_{n-1,k-1}(a_n))| \geq |F_e \cap E(S_{n-1,k-1}(a_{n-1}))| \geq \dots \geq |F_e \cap E(S_{n-1,k-1}(a_1))|$ for $1 \leq a_i \leq n$ and $1 \leq i \leq n$. Then at least three $S_{n-1,k-1}(a_j)$ s are fault-free in the $S_{n,k}$ with $|F_e| \leq n-3$, where $1 \leq a_j \leq n$ and $1 \leq j \leq 3$. Let $N = |V(S_{n-1,k-1})|$. According to Theorem 2, a fault-free $S_{n-1,k-1}(a_j)$ has a cycle of length ranged from 3 to N and a path between two endvertices of a 1-edge with l vertices for $2 \leq l \leq N$ in the $S_{n-1,k-1}(a_j)$, where $n \geq 6$ and $k \geq 3$.

Lemma 9. An $S_{n,k}$ is $(n-3)$ -edge-fault-tolerant pancyclic for $3 \leq k \leq n-4$ and $n \geq 7$.

Proof. A cycle of length ranged from $N+1$ to nN is built step by step and two cases have to be considered as follows.

Case 1. $|F_e \cap E(S_{n-1,k-1}(a_n))| < n-4$. Applying Algorithm 1 in the $S_{n,k}$, there exists a fault-free cycle C of length 6. Let $p = p_1 p_2 \dots p_k$ in C be the selected vertex in Step 1 of Algorithm 1, such that $S_{n-1,k-1}(p_1)$ and $S_{n-1,k-1}(p_k)$ are fault-free.

1. Let $C = \langle p, p^2, p^3, p^4, p^5, p^6, p \rangle$, $p = p_1 p_2 \dots p_k$, and $\alpha \in \langle n \rangle - \{p_q \mid 1 \leq q \leq k\}$, such that $(p, p^6) \in E(S_{n-1,k-1}(p_k))$, $(p^2, p^3) \in E(S_{n-1,k-1}(p_1))$ and $(p^4, p^5) \in E(S_{n-1,k-1}(\alpha))$, respectively. Build a path with $N-3$ vertices between p and p^6 in the

$S_{n-1,k-1}(p_k)$. A cycle of length $N+1$ can be constructed as shown in Figure 8.

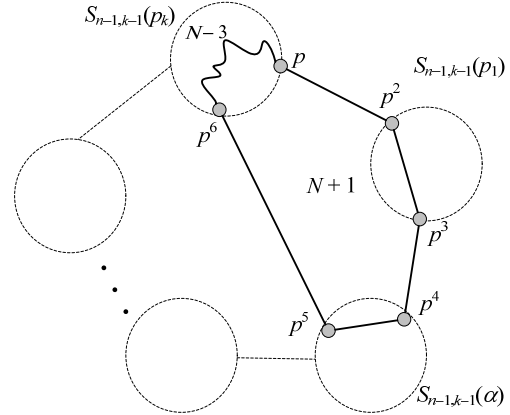


Figure 8. A cycle of length $N+1$ in an $S_{n,k}$.

2. By adding vertices in the $S_{n-1,k-1}(p_k)$, which are excluded by the cycle constructed in Step 1, into the cycle one-by-one, a cycle of length ranged from $N+2$ to $N+4$ can be established.
3. For constructing a cycle of length $N+5$, adopt C in Step 1 as a basic cycle again, build a path with 3 vertices between p and p^6 in the $S_{n-1,k-1}(p_k)$, and construct a Hamiltonian path in the $S_{n-1,k-1}(\alpha)$ between p^4 and p^5 according to Lemma 1. Thus, a cycle of length $N+5$ can be obtained shown in Figure 9.

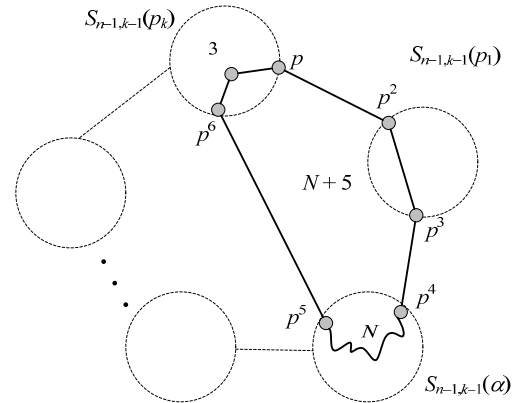


Figure 9. A cycle of length $N+5$ in an $S_{n,k}$.

4. By adding vertices in $S_{n-1,k-1}(p_k)$ and $S_{n-1,k-1}(p_1)$, which are excluded by the cycle constructed in Step 3, into the cycle one-by-one, a cycle of length ranged from $N+6$ to $3N$ can be established.
5. For constructing a cycle of length $3N+1$, let $\gamma_1 \in \langle n \rangle - \{p_1, p_k, \alpha\}$, applying Algorithm 2 in the $S_{n,k}$, there exists a fault-free path $\langle q^1, q^2, q^3, q^4 \rangle$ with 4 vertices crossing $S_{n-1,k-1}(p_k)$, $S_{n-1,k-1}(\gamma_1)$ and $S_{n-1,k-1}(\alpha)$, such that $q^1 \in V(S_{n-1,k-1}(p_k))$ and $q^2, q^3 \in V(S_{n-1,k-1}(\gamma_1))$, $q^4 \in V(S_{n-1,k-1}(\alpha))$. Build a path with $N-1$ vertices between p^2 and p^3 in

the $S_{n-1,k-1}(p_1)$, a Hamiltonian path in the $S_{n-1,k-1}(p_k)$ between p and q^1 , and a Hamiltonian path in the $S_{n-1,k-1}(\alpha)$ between p^4 and q^4 according to Lemma 1. Thus, a cycle of length $3N+1$ can be obtained as shown in Figure 10. Next, reconstructing a Hamiltonian path in $S_{n-1,k-1}(p_1)$ between p^2 and p^3 , a cycle of length $3N+2$ can be obtained.

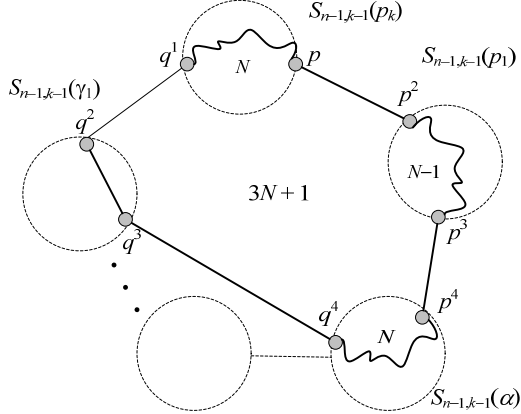


Figure 10. A cycle of length $3N+1$ in an $S_{n,k}$.

6. For constructing a cycle of length $3N+3$, first adopt the cycle constructed by Step 5 as the basic cycle. Then rebuild a path with 3 vertices between p^2 and p^3 in the $S_{n-1,k-1}(p_1)$, and reconstruct a Hamiltonian path in the $S_{n-1,k-1}(\gamma_1)$ between q^2 and q^3 according to Lemma 1. Thus, a cycle of length $3N+3$ can be obtained as shown in Figure 11.

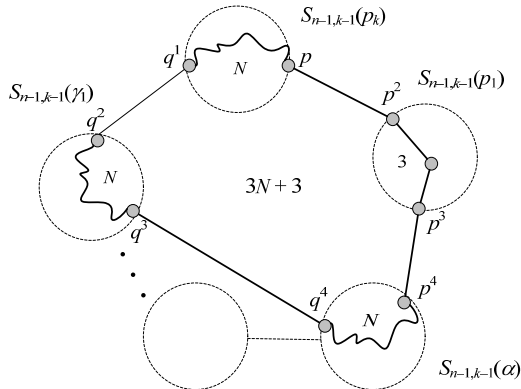


Figure 11. A cycle of length $3N+3$ in an $S_{n,k}$.

7. By adding vertices in the $S_{n-1,k-1}(p_1)$, which are excluded by the cycle constructed in Step 6, into the cycle one-by-one, a cycle of length ranged from $3N+4$ to $4N$ can be established.
8. Similar to Step 5, let $\gamma_1, \gamma_2 \in \langle n \rangle - \{p_1, p_k, \alpha\}$, applying Algorithm 2 in the $S_{n,k}$, there exists a fault free path $\langle q^1, q^2, q^3, q^4, q^5, q^6 \rangle$ with $q^1 \in V(S_{n-1,k-1}(p_k))$ and $q^2, q^3 \in V(S_{n-1,k-1}(\gamma_1))$, $q^4, q^5 \in V(S_{n-1,k-1}(\gamma_2))$, and $q^6 \in V(S_{n-1,k-1}(\alpha))$. For constructing a cycle of length $4N+1$, build a

path with $N-1$ vertices between p^2 and p^3 in the $S_{n-1,k-1}(p_1)$, a Hamiltonian path in each of $S_{n-1,k-1}(p_k)$, $S_{n-1,k-1}(\gamma_1)$ and $S_{n-1,k-1}(\alpha)$ between p and q^1 , q^2 and q^3 , and p^4 and q^6 , respectively. Thus, a cycle of length $4N+1$ can be obtained as shown in Figure 12. Additionally, reconstruct a Hamiltonian path in the $S_{n-1,k-1}(p_1)$ between p^2 and p^3 , a cycle of length $4N+2$ can be obtained. According to Steps 6 and 7, a cycle of length ranged from $4N+3$ to $5N$ can be established.

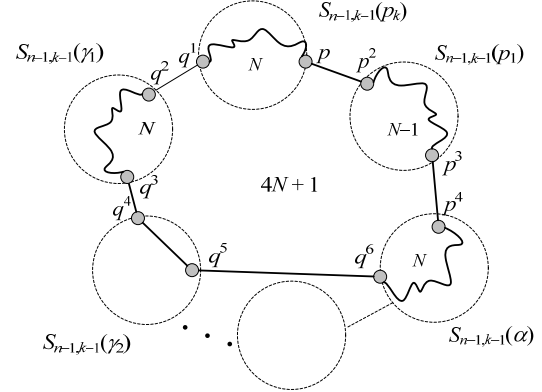
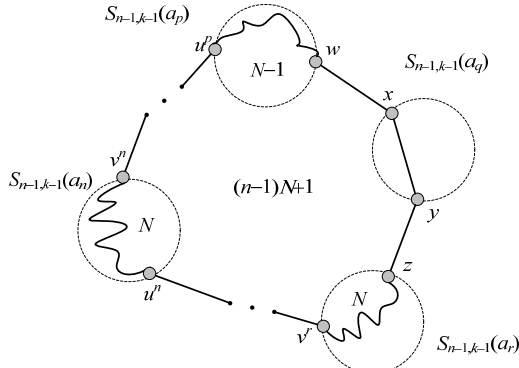


Figure 12. A cycle of length $4N+1$ in an $S_{n,k}$.

9. Referring to Steps 5 to 8, a cycle of length ranged from $5N+1$ to nN can be built.

Case 2. $|F_e \cap E(S_{n-1,k-1}(a_n))| \geq n-4$. Similar to Case 1, a cycle of length ranged from $N+1$ to $(n-1)N$ can be built and every vertex in the constructed cycle is not in the $S_{n-1,k-1}(a_n)$. According to Lemma 1, there exists a Hamiltonian cycle HC in $S_{n,k}$. Then vertices $u^i, v^i \in V(S_{n-1,k-1}(a_i))$ and each subpath $\langle u^i, \dots, v^i \rangle$ of the HC is a Hamiltonian path in the $S_{n-1,k-1}(a_i)$, where $1 \leq i \leq n$. Moreover, there exist three contiguous subpaths $\langle u^p, \dots, v^p \rangle$, $\langle u^q, \dots, v^q \rangle$ and $\langle u^r, \dots, v^r \rangle$ in the HC, such that $S_{n-1,k-1}(a_p)$ and $S_{n-1,k-1}(a_q)$ are fault-free and $r \neq n$, where $1 \leq p \neq q \neq r < n$. A fault-free path with 4 vertices can be built as $\langle w, x, y, z \rangle$ crossing $S_{n-1,k-1}(a_p)$, $S_{n-1,k-1}(a_q)$, $S_{n-1,k-1}(a_r)$, such that (x, y) is a 1-edge, (w, x) and (y, z) are k -edges. A cycle of length ranged from $(n-1)N+1$ to nN is built step by step as follows.

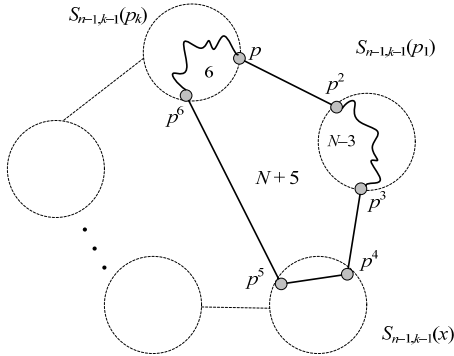
1. According to Lemma 1, construct a Hamiltonian path between u^p and w in the $S_{n-1,k-1}(a_p) \setminus t$, where $t \in V(S_{n-1,k-1}(a_p)) - \{u^p, w\}$, and a Hamiltonian path of $S_{n-1,k-1}(a_r)$ between y and z . Figure 13 depicts the cycle of length $(n-1)N+1$.
2. By adding vertices in $S_{n-1,k-1}(a_p)$ and $S_{n-1,k-1}(a_q)$, which are excluded by the cycle constructed in Step 1, into the cycle one-by-one, a cycle of length ranged from $(n-1)N+2$ to nN can be established. $\square$


 Figure 13. A cycle of length $(n-1)N+1$ in an $S_{n,k}$.

Lemma 10. An $S_{n,n-2}$ ($S_{n,n-3}$) is $(n-3)$ -edge-fault-tolerant 6-weak-pancyclic, where $n \geq 6$.

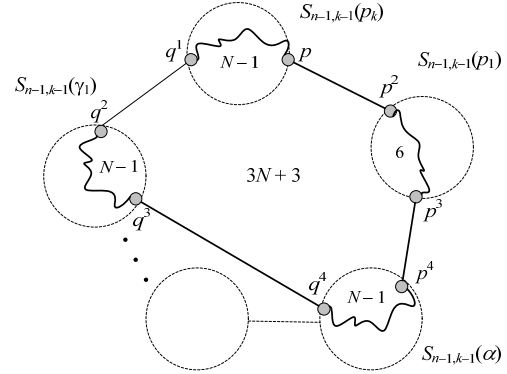
Proof. This proof is very similar to that of Lemma 9. Because an $S_{n,n-2}$ ($S_{n,n-3}$) has no cycles of length 4, 5 (5), some constructions should be modified as follows. Let $k = n-2$ (or $k = n-3$) for the $S_{n,k}$.

Case 1. $|F_e \cap E(S_{n-1,k-1}(a_n))| < n-4$. Constructions of cycles of lengths $N+5$ and $lN+m$ in the $S_{n,k}$ should be modified for $3 \leq l \leq n-1$, $3 \leq m \leq 5$. For building a cycle of length $N+5$, referring to the cycle in Figure 9, rebuild a path with 6 vertices between p and p^6 in the $S_{n-1,k-1}(p_k)$, and a path with $N-3$ vertices between p^2 and p^3 in the $S_{n-1,k-1}(p_1)$. Thus, a cycle of length $N+5$ can be obtained as shown in Figure 14.

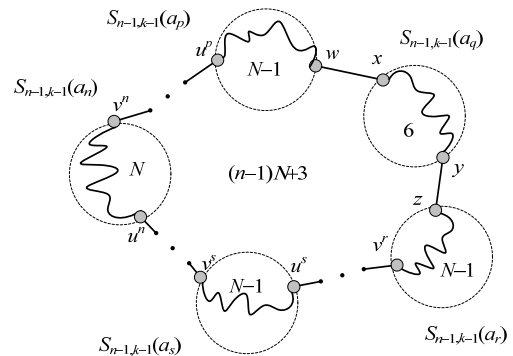

 Figure 14. A cycle of length $N+5$ in an $S_{n,k}$.

Since constructing a cycle of length $lN+m$ is similar to each other for $3 \leq l \leq n-1$, $3 \leq m \leq 5$, only the construction of a cycle of length ranged from $3N+3$ to $3N+5$. Referring to the cycle in Figure 11, rebuild a path with 6 vertices between p^2 and p^3 in the $S_{n-1,k-1}(p_1)$ and a Hamiltonian path between p and q^1 in the $S_{n-1,k-1}(p_k) \setminus t^1$ according to Lemma 1, where $t^1 \in V(S_{n-1,k-1}(p_k)) - \{p, q^1\}$. If $|F_e \cap E(S_{n-1,k-1}(\gamma_1))| = 0$, a Hamiltonian path between q^2 and q^3 in the $S_{n-1,k-1}(\gamma_1) \setminus t^2$ can be established, where $t^2 \in V(S_{n-1,k-1}(\gamma_1)) - \{q^2, q^3\}$. Otherwise, assume $(s, s') \in F_e \cap E(S_{n-1,k-1}(\gamma_1))$, since $|F_e \cap E(S_{n-1,k-1}(\gamma_1)) - \{s, s'\}| + |\{s\}| \leq n-5$, a

Hamiltonian path can be built between q^2 and q^3 in the $S_{n-1,k-1}(\gamma_1)$ s according to Lemma 1. For the same reason, a path with $N-1$ vertices can be constructed between q^4 and p^4 in the $S_{n-1,k-1}(\alpha)$. Thus, a cycle of length $3N+3$ can be obtained as shown in Figure 15. By adding vertices in $S_{n-1,k-1}(p_k)$ and $S_{n-1,k-1}(x)$ into the cycle one-by-one, cycles of lengths $3N+4$ and $3N+5$ can be established.


 Figure 15. A cycle of length $3N+3$ in an $S_{n,k}$.

Case 2. $|F_e \cap E(S_{n-1,k-1}(a_n))| \geq n-4$. The construction of a cycle of length ranged from $(n-1)N+3$ to $(n-1)N+5$ should be modified. Referring to the cycle in Figure 13, let subpath $\langle u^s, \dots, s^s \rangle$ of the HC be a Hamiltonian path in the $S_{n-1,k-1}(a_s)$, where $s \neq n$. Refer to Case 1 to reconstruct a path with $N-1$ vertices in each of $S_{n-1,k-1}(a_p)$, $S_{n-1,k-1}(a_r)$ and $S_{n-1,k-1}(a_s)$ between u^p and w , z and v^r , and u^s and v^s , respectively. Rebuild a path with 6 vertices between x and y . Thus, a cycle of length $(n-1)N+3$ can be obtained as shown in Figure 16. By adding vertices in $S_{n-1,k-1}(a_r)$ and $S_{n-1,k-1}(a_s)$ into the cycle one-by-one, cycles of lengths $(n-1)N+4$ and $(n-1)N+5$ can be established. $\square$


 Figure 16. A cycle of length $(n-1)N+3$ in an $S_{n,k}$.

Lemma 11. An $S_{5,3}$ is 2-edge-fault-tolerant 6-weak-pancyclic.

Proof. Without loss of generality, assume that $|F_e \cap E(S_{4,2}(a_5))| \geq |F_e \cap E(S_{4,2}(a_4))| \geq \dots \geq$

$|F_e \cap E(S_{4,2}(a_1))|$. Hence, at least three $S_{4,2}(a_i)$'s are fault-free in the $S_{5,3}$ with $|F_e| \leq 2$, where $1 \leq a_i \leq 5$ and $1 \leq i \leq 5$. By exhaustive search, an $S_{4,2}$ is 6-weak-panconnected. Thus, a cycle of length ranged from 6 to 12 can be constructed in an $S_{4,2}(a_j)$, where $1 \leq a_j \leq 5$ and $1 \leq j \leq 3$. A cycle of length ranged from 13 to 60 is built step by step as follows.

1. Construct a fault-free cycle $\langle u^1, v^1, u^2, v^2, u^3, v^3, u^1 \rangle$ with $(u^i, v^i) \in E(S_{n-1,1}(a_j))$ for $1 \leq j \leq 3$. Building a path with 9 vertices between u^1 and v^1 in the $S_{4,2}(a_1)$, a cycle of length 13 can be obtained as shown in Figure 17.

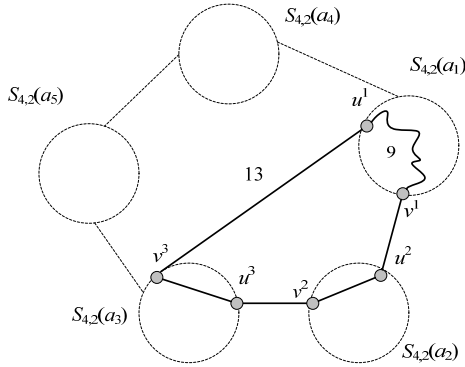


Figure 17. A cycle of length 13 in an $S_{5,3}$.

2. Since an $S_{4,2}$ is 6-weak-panconnected, a path with m vertices for $6 \leq m \leq 12$ can be constructed in an $S_{4,2}(a_j)$, where $1 \leq j \leq 3$. Hence, a cycle of length ranged from 14 to 36 can be obtained.
3. According to Lemma 1, there exists a Hamiltonian cycle HC in the $S_{5,3}$. Then, vertices $u^i, v^i \in V(S_{4,2}(a_i))$ and subpath $\langle u^i, \dots, v^i \rangle$ of the HC is a Hamiltonian path in the $S_{4,2}(a_i)$, where $1 \leq i \leq 5$. Moreover, without loss of generality, assume that there exist two contiguous subpaths $\langle u^1, \dots, v^1 \rangle$ and $\langle u^2, \dots, v^2 \rangle$ in the HC. Rebuild a fault free path as $\langle u^1, x, y, z, v^2 \rangle$, such that $x, y \in V(S_{4,2}(a_1))$ and $z \in V(S_{4,2}(a_2))$, reconstruct a path with 8 vertices between u^3 and v^3 in the $S_{4,2}(a_3)$. Thus, a cycle of length 37 can be obtained.
4. Since a path with m vertices for $6 \leq m \leq 12$ can be constructed in an $S_{4,2}(a_j)$, where $1 \leq j \leq 3$, a cycle of length ranged from 38 to 60 can be similarly obtained. $\square$

Based on the results provided by Li [18] and Xu et al. [24] and combining Lemmas 3, 4, 5, 9, 10 and 11, the following theorem can be obtained.

Theorem 12. An $S_{n,k}$ is $(n-3)$ -edge-fault-tolerant pancyclic if $k = 1$ and $n \geq 4$, $k = 2$ and $n \geq 6$, or $3 \leq k \leq n-4$ and $n \geq 7$; an (n, k) -star graph is $(n-3)$ -edge-fault-tolerant 6-weak-pancyclic, if $n-3 \leq k \leq n-2$ and $n \geq 5$, and an (n, k) -star graph is

$(n-3)$ -edge-fault-tolerant 6-weak-edge-bipancyclic, if $k = n-1$ and $n \geq 4$.

4 Conclusions

This work reveals that an (n, k) -star graph is $(n-3)$ -edge-fault-tolerant pancyclic if $k = 1$ and $n \geq 4$, $k = 2$ and $n \geq 6$, or $3 \leq k \leq n-4$ and $n \geq 7$; an (n, k) -star graph is $(n-3)$ -edge-fault-tolerant 6-weak-pancyclic, if $n-3 \leq k \leq n-2$ and $n \geq 5$, and an (n, k) -star graph is $(n-3)$ -edge-fault-tolerant 6-weak-edge-bipancyclic, if $k = n-1$ and $n \geq 4$. Moreover, if there are $n-2$ faulty edges incident to some vertex in an (n, k) -star graph, then the faulty (n, k) -star graph is not Hamiltonian (or contains no cycle of length $|V(S_{n,k})|$). Thus, the result of this work is optimal with respect to the number of edge faults tolerated.

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